Foundations of Generative AI (ICT4403)
Assignment 2

1. Select a suitable textual dataset of your choice and develop an autoregressive text-generation system using Simple RNN, Bidirectional RNN, LSTM, and GRU models. Train all four models under comparable experimental conditions to predict the next word and generate text sequentially from suitable seed sequences.

   Tasks:

   i. For the selected dataset, describe its characteristics, preprocess the text, and construct the input–target sequences required for autoregressive modelling. Implement and train all four models using the prepared dataset.

   ii. Evaluate and compare the models using suitable quantitative measures, such as loss, perplexity, prediction accuracy, and training time, as well as qualitative characteristics of the generated text, including coherence, diversity, contextual consistency, and repetition.

   iii. Generate text from the same seed sequences using at least three sampling-temperature values. Analyse the influence of temperature on coherence, diversity, and repetition, and justify which model provides the most effective autoregressive generation for the selected dataset.

Submission Procedure

Step 1: Complete the Assignment

- Complete all the **implementation, evaluation, comparison, and analysis** as specified in the question.
- Ensure that the program executes correctly and produces the reported results.

Step 2: Upload Your Work to GitHub

1. Create a **GitHub repository** with a clear and relevant name. If it is private, provide access to the evaluator.
2. Upload the implementation as a **Jupyter Notebook (.ipynb)** or **Python source files (.py)**, along with the required supporting files, results, plots, and generated-text samples.
3. Provide the correct **source link** for the textual dataset used.
4. Include a **README.md** file containing:
   - problem statement and objective,
   - dataset description and source link,
   - brief description of the models implemented,
   - execution steps and required dependencies,
   - results obtained,
   - model comparison and evaluation,
   - sampling-temperature analysis, and

- conclusion identifying the best-performing model.

5. Organize the repository properly and ensure that all files and links are accessible.

Step 3: Share the GitHub Repository Link

1. Create a **Word document** named: YourName_RegNum_FGAI_Assignment2.docx
2. Include the following details:
    - student name,
    - registration number,
    - assignment title,
    - title of the developed autoregressive text-generation system,
    - name and link of the dataset used, and
    - GitHub repository link.

Step 4: Convert to PDF

- Convert the Word document to **PDF format** using File → Save As → PDF.

Step 5: Upload to LMS

- Upload the final **PDF file** to the specified assignment submission section in the **Learning Management System (LMS)** before the deadline.